



URL Shortener

Project Documentation

A Full-Stack Web Application with Analytics

Project:	URL Shortener
Stack:	Spring Boot 3 · React 18 · MySQL 8
Language:	English
Date:	05 June 2026

1. Project Overview

The URL Shortener is a full-stack web application that allows users to convert long, complex URLs into short, shareable links. Every time a short link is clicked, the system records the event and provides analytics to the user.

Live URL

Frontend !' <http://localhost:5173> | Backend API !' <http://localhost:8080>

% Core Features

- Shorten any valid HTTP/HTTPS URL into a 6-character unique code
- Automatic redirect from short URL to the original destination
- Click tracking — every visit is recorded in the database
- Per-URL analytics with a daily click chart (last 30 days)
- Global statistics dashboard showing platform-wide trends
- Paginated list of all shortened URLs (10 per page)
- Copy to clipboard and direct open buttons for each URL
- Swagger / OpenAPI documentation at `/swagger-ui.html`

2. Technology Stack

% Backend

- Java 21 + Spring Boot 3.2.0
- Spring Data JPA + Hibernate (ORM)
- MySQL 8 (database)
- Lombok (reduce boilerplate code)
- SpringDoc OpenAPI (auto-generated Swagger docs)
- Maven (build tool)

% Frontend

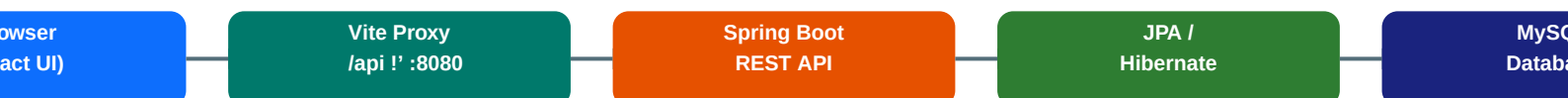
- React 18 + TypeScript
- Vite 5 (build tool / dev server)
- Axios (HTTP client)
- Recharts (interactive charts)
- Vanilla CSS (custom styling)

% Database

- MySQL 8 running locally on port 3306
- Database name: `url_shortener`
- Two tables: `urls` and `click_events`

3. System Architecture

The application follows a classic 3-tier architecture:



% Layer Responsibilities

- Presentation Layer ! React components (HelloBANNER, RecentUrls, Statistics, Analytics Modal)

- API Layer

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4. Database Design

% Table: urls

Column	Type	Description
id	BIGINT AUTO_INCREMENT	Primary key
original_url	TEXT NOT NULL	The long original URL
short_code	VARCHAR(20) UNIQUE	6-char random code (e.g. aB3kZ9)
created_at	TIMESTAMP	When the short URL was created
click_count	INT DEFAULT 0	Total number of clicks
last_accessed_at	TIMESTAMP NULL	Last time the short URL was clicked

% Table: click_events

Column	Type	Description
id	BIGINT AUTO_INCREMENT	Primary key
url_id	BIGINT (FK)	References urls.id (CASCADE DELETE)
clicked_at	TIMESTAMP	Exact timestamp of each click

Design Note

Each click is stored as a separate row in click_events. This allows accurate daily grouping via DATE(clicked_at) in SQL queries — supporting both per-URL analytics and global statistics.

5. REST API Endpoints

Method + Path	Purpose	Response
POST /api/urls	Create a new short URL	JSON / 302 Redirect
GET /api/urls?page=0&size=10	List all URLs (paginated)	JSON / 302 Redirect
GET /api/urls/{id}/analytics	Get per-URL click analytics	JSON / 302 Redirect
GET /api/urls/stats	Get global 14-day statistics	JSON / 302 Redirect
GET /{shortCode}	Redirect to original URL (302)	JSON / 302 Redirect

% Example — Create Short URL

```
POST /api/urls
Content-Type: application/json

Request:  { "originalUrl": "https://www.example.com/very/long/path" }

Response: {
  "id": 1,
  "originalUrl": "https://www.example.com/very/long/path",
  "shortCode": "aB3kZ9",
  "shortUrl": "http://localhost:8080/aB3kZ9",
  "clickCount": 0,
  "createdAt": "2026-06-05T10:39:00"
}
```

% Short Code Generation Logic

The service generates a random 6-character code using the alphabet [A-Z a-z 0-9] (62 chars), giving 62^6 = 56 billion possible combinations. `SecureRandom` is used for security. If a collision occurs, the system retries up to 10 times before throwing an error.

6. Frontend Components

Component	Role
Header.tsx	Top navigation bar with the brand title
HeroBanner.tsx	URL input form — validates, calls <code>POST /api/url</code> , shows errors
RecentUrls.tsx	Paginated table of all URLs with <code>Copy</code> , <code>Open</code> , <code>Analytics</code> buttons
AnalyticsModal.tsx	Modal popup showing per-URL daily click chart (<code>AreaChart</code>)
Statistics.tsx	Global chart — <code>ComposedChart</code> (<code>Area</code> + <code>Bar</code>) showing 14-day
Pagination.tsx	Page navigation component

% API Proxy Configuration (`vite.config.ts`)

Vite proxies all `/api/*` requests to `http://localhost:8080`, allowing the frontend to call the backend without CORS issues during development.

```
proxy: {  
  '/api': { target: 'http://localhost:8080', changeOrigin: true }  
}
```

% State Management

- React `useState` — local component state (URLs list, loading, errors)
- React `useCallback` — memoized `fetchUrls` to avoid unnecessary re-renders
- React `useEffect` — triggers data fetch on page number change
- Optimistic UI — new URL is prepended instantly before server confirms

7. How to Run the Project

%_ Step 1 — Set up MySQL Database

```
-- Run in MySQL client:
source d:/url-shorterner/database/schema.sql

-- Optional test data:
source d:/url-shorterner/database/seed.sql
```

%_ Step 2 — Configure Password

```
# File: backend/src/main/resources/application.properties
spring.datasource.username=root
spring.datasource.password=root    !• update if different
```

%_ Step 3 — Start Backend

```
cd d:\url-shorterner\backend
$env:JAVA_HOME = "C:\Users\udhay\.antigravity-ide\extensions\redhat.java-1.54.0-win32-x64\jre\21.0.10-win
$env:PATH = "$env:JAVA_HOME\bin;$env:PATH"
d:\url-shorterner\maven\apache-maven-3.9.6\bin\mvn.cmd spring-boot:run

!' Backend starts at http://localhost:8080
```

%_ Step 4 — Start Frontend

```
cd d:\url-shorterner\frontend
npm install      (first time only)
npm run dev

!' Frontend starts at http://localhost:5173
```

Swagger UI

API documentation available at <http://localhost:8080/swagger-ui.html>

8. Project Folder Structure

```
url-shortener/  
% % % backend/                                !• Spring Boot application  
%   % % % src/main/java/com/urlshortener/  
%       % % % controller/                    !• UrlController, RedirectController  
%       % % % service/                      !• UrlService, UrlServiceImpl  
%       % % % repository/                   !• UrlRepository, ClickEventRepository  
%       % % % model/                        !• Url.java, ClickEvent.java  
%       % % % dto/                          !• Request/Response objects  
%       % % % exception/                    !• Custom exceptions  
%       % % % mapper/                       !• UrlMapper  
%   % % % src/main/resources/  
%       % % % application.properties  
% % % frontend/                              !• React + Vite application  
%   % % % src/  
%       % % % components/                   !• UI components  
%       % % % services/api.ts               !• Axios API calls  
%       % % % types/index.ts                !• TypeScript interfaces  
% % % database/  
%   % % % schema.sql                       !• Table definitions  
%   % % % seed.sql                         !• Sample data  
% % % maven/                               !• Bundled Maven 3.9.6
```